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Optimizing rooftop photovoltaic distributed generation with battery storage for peer-to-peer energy trading

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HIGHLIGHTS

- Optimization model developed for PV–battery systems in P2P energy trading market.
- P2P energy trading implements real-world constraints and market signals.
- Analysis shows household energy savings sensitive to multiple parameters.
- Maximal savings up to 28% achieved when equipped with large PV-battery on weekdays.
- Battery-only households achieve fewer savings compared to those without any DG.

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ABSTRACT

Distributed generation (DG) based on rooftop photovoltaic (PV) systems with battery storages is a promising alternative energy generation technology to reduce global greenhouse gas emissions. As regulatory tariff-based incentives are diminishing, innovative solutions are required to sustain this renewable energy generation. An optimization model is proposed to maximize the economic benefits for rooftop PV-battery DG in a peer-to-peer (P2P) energy trading environment. The goal of the proposed model is to investigate the feasibility of such renewable source participated P2P energy trading by examining the economic benefits. The model is illustrated in a simulation framework for a local community with 500 households under real-world constraints encompassing PV systems, battery storage, customer demand profiles and market signals including the retail price, feed-in tariff and P2P energy trading mechanism. Interactions among peer-to-peer trading stakeholders are examined, quantifying household savings for different scenarios of this P2P-based DG. Household energy savings are identified to be sensitive to many factors including the scale of PV systems, the PV penetration, the P2P trading margins, the presence of battery storage and energy trading time. The model shows that maximal savings up to 28% can be achieved by households equipped with larger PV systems and battery storages during weekdays from an exemplified case. The sensitivity analysis demonstrates that households with PV systems have lower savings when PV penetration is high owing to excessive energy traded on the P2P market, pushing down the clearing price and the savings gain. Households with a battery-only configuration are shown to achieve fewer savings in contrast to households without any renewable resources in a P2P trading community. The model's energy insights are the beginning of understanding the actual impact of policy, market and technical signals on economic benefits for household distributed renewable generation in a P2P energy trading market.

1. Introduction

The global energy market is undergoing drastic changes with an increasing consumer appetite for renewable resources and battery storage to reduce greenhouse gas emissions. Australia is a world leader in the penetration of household solar photovoltaic (PV) panels with 15% of its households (around 1.4 million) having roof-mounted PV in 2015, capturing 4.5 GW of solar peak capacity [1]. Morgan Stanley estimates the Australian market for household battery storage to hit one million by 2020 [2]. Rooftop PV generation is projected to increase more than sixfold by 2050 in Australia [3]. However, the demand side participation of renewable generation in Australia (i.e., with solar PV) is heavily subsidized by governments leaving it an uneconomical alternative to traditional fossil fuel-based generation. As tariff-based incentives are

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now diminishing, it is imperative to develop innovative solutions to sustain installation of renewable energy generation. A recent techno-economic analysis [4] shows that the viability of residential renewable energy generation systems (more specifically, PV-battery systems) is significantly dependent on regulatory subsidies [5] and cost reductions. This analysis also highlights the need to develop intelligent control strategies to optimize the flow of energy under realistic system parameters [4]. Developing optimization models for rooftop PV-battery distributed generation and investigating new market frameworks, such as peer-to-peer (P2P) energy trading market to maximize benefits are reasonable steps to proceed.

Distributed generation (DG) is defined as electric power generation within a distribution network or on the demand side of the customer network [6]. DG enables close to load generation with flexible demand response management and high penetration of renewable generation [7]; therefore DG is claimed as a more sustainable electricity generation alternative than the conventional fossil-fuel dependent centralized generation paradigm [8]. Distributed energy technologies such as energy storage have been developed and applied to further embrace the advantages of PV systems [9]. To efficiently utilize these energy resources for minimizing the energy supply cost or for maximizing the grid energy export profit, many studies have focused on building complex optimization models to determine optimal schedules of energy generated from different energy resources [10]. Similar optimization problems have been investigated for utility companies [11] or embedded power systems (e.g., for office buildings [12], industrial facilities [13], or a PV generating plant [14]) but are not within the scope of our paper.

The optimization studies related to DG for residential households can be classified into two main categories: (a) optimization models for operational decisions related to a residential customer or an off-grid system and (b) centralized optimization models to determine the optimal operational decisions for a number of consumers (e.g., 2–4 households) [15,16]. In the first category, a leading research direction is an optimal DG system design (mainly for a residential customer) consisting of many key components such as a PV system and battery storage to achieve the highest financial value over the planning horizon of the system [17–19]. In these studies, the researchers aimed at generating an optimal energy dispatching schedule based on the available distributed energy resources. Most proposed optimization problems have been formulated as linear programming (LP) models, which can be solved efficiently using standard LP solvers. Depending on the goal of each study, the planning horizon can be adjusted (from 1 day to 1 year). For example, Bordin et al. [18] developed a linear programming model to investigate how different operational patterns of an off-grid power system impacts the degradation costs of batteries. Their goal is to provide a guideline for designers on choosing a battery for off-grid sites based on battery configurations and diesel costs. Hassan et al. [19] developed a mixed integer linear programming (MILP) model to analyze the benefits while considering different electricity tariffs and battery storage to maximize feed-in tariff revenue streams for the existing PV generating system. There is a constraint to avoid buying and selling electricity simultaneously at each time step. The MILP is simple but applied to a very long planning horizon. In another study, Linssen et al. [17] used simulation-based optimization to take into consideration temporal high-resolution consumer load and PV production profiles, technical and economical PV and battery system parameters as well as the regulatory framework in Germany. Another main research direction in the first category is optimizing the operational decisions that take advantage of multiple energy resources to satisfy energy demand based on a predefined system configuration [20–22]. For example, Hill and Nwankpa [20] develop an LP model to optimize energy resources used to satisfy the demand of a specific customer with a PV system and a battery (similar but much simpler than our model). Their model took the battery constraints (capacity and charging rate) into account and assumed that the cost for using a PV system and battery is zero. However, the retail price is simplified, and the feed-in tariff is not considered in their study. Iwafune et al. [21] developed a home energy management system that controls a residential battery system connected to a rooftop PV system taking into account the energy load and PV generation forecast errors. The proposed system performs scheduling 24-h ahead of time and executes the planned battery operation on the target day. The goal is to minimize consumer cost and optimize battery operation. Simplified storage and discharge modes of the battery are also modeled in the paper. The maximum charging/discharging rate is not considered (i.e., the model is bounded merely by the battery power capacity).

For the second category, researchers have mainly focused on demand-side management (DSM) for a few households. Alam et al. [15] developed a sophisticated non-linear optimization model that optimizes smart device operations. However, this model is only tested on two households. Examining the smart home context, Joo and Choi [16] formulated a MILP model for scheduling the energy consumption of multiple smart homes with distributed energy resources that minimize the total energy cost and discomfort cost (caused by the deviation of a consumer's preferred temperature from the indoor temperature). A two-level optimization algorithm was also proposed to solve this problem. Owing to the complexity of the model, only four simulated households are tested. While these studies contributed to proving the feasibility and benefits of optimization techniques in DG with PV and battery storage, little research has occurred in the scope of a P2P energy trading market consisting of multiple households with renewable energy resources.

P2P energy trading enables consumers to become prosumers by selling their surplus energy from their renewable resources and storage to their neighborhood. P2P energy trading thus empowers prosumers to trade renewable energy beyond geographic boundaries. P2P energy trading start-ups have successfully attracted millions of dollars for investments to run field trials in Australia [23], the UK [24] and Europe [25]. Many other projects and trials are mentioned in [26]. These trials are at an early stage of developing P2P energy trading platforms covering preliminary issues including (a) the development of business models and system level design [27], (b) the provision of information and communication technologies (ICT) to connect stakeholders, (c) the development of blockchain technologies to simplify metering and billing systems [26] and (d) the provision of a transactional layer for renewable assets [28]. Little attention has been paid to optimizing operational decisions related to energy resources, e.g., PV systems and energy storage. While the P2P energy market aims to offer more flexibility for consumers, it requires techniques to handle complexity in their decision-making processes, i.e., where should the surplus energy be exported to or when should the stored energy be used. These daily operational decisions and the consumers' behavior will govern their resource utilization and strongly influence their energy costs.

There has been a limited number of optimization studies in the context of P2P energy trading. Alam et al. [15] developed a mixed integer non-linear programming (MINLP) model to integrate DSM with P2P energy trading for smart homes. Unfortunately, there is no trading mechanism considered in this paper and the simulation is only performed for two households, which is not sufficient to demonstrate any benefit of P2P trading. Munsing et al. [29] proposed an interesting theoretical optimization model for a P2P energy market based on blockchains in a microgrid network. They also proposed an alternating direction method of multipliers (ADMM) technique to find an optimal schedule for a mix of batteries, shapable loads, and deferrable loads. Their simulation (based on a theoretical network model) showed the advantages of P2P energy trading when operational costs related to energy resources are optimized. Although the results are promising, there are some limitations to their approach. In particular, their study does not investigate how the size of distributed energy resources, PV penetration, market signals (e.g., the feed-in tariff and P2P trading margin) and customer energy consumption preference will influence the energy costs of households and the cost-benefit for investment in

DG technology, especially within residential households. Furthermore, they suggested the communication overhead required for ADMM can limit the application of their proposed approach to day-ahead scheduling scenarios while real-time control needs to be handled by a conventional centralized approach.

The main contribution of this paper is the development of an optimization model for rooftop PV with battery storage in the context of P2P energy trading. This study proposes a mixed integer linear programming (MILP) model to optimize the operational decisions of a large number of households participating in a P2P trading electricity market. Contrary to existing optimization studies, the proposed model considers a peer-to-peer energy trading mechanism and practical constraints on DG resources, especially PV systems and battery storage. Also, the model considers a group of heterogeneous households and their interactions instead of a single residential household or a power system commonly examined in the existing studies. As compared to non-linear optimization models developed in Alam et al. [15] and Munsing et al. [29], the proposed MILP model can explicitly incorporate a pricing mechanism, and it is efficient enough to cope with a large number of households in real-world scenarios. The proposed model can be used to assist various stakeholders in investigating the feasibility and benefits of peer-to-peer renewable energy trading. Furthermore, the model allows us to understand the techno-economic trade-offs related to various operational factors including tariffs, households renewable investment and saving in renewable DG under different scenarios.

This paper is organized as follows: Section 2 presents the proposed optimization model for P2P energy trading. Section 3 presents case scenarios for optimized energy flow management within a P2P energy trading simulation reflecting real-world conditions. The results are discussed in Section 4. Section 5 concludes the paper with future works.

2. Mixed integer linear programming model for P2P energy trading

Fig. 1 shows the conceptual model of the P2P energy trading market considered in this study. Under this setting, the distributed consumers can save their energy cost by participating in P2P trading, which results in an energy value payment by the distributed consumer to the distributed generator via an aggregator. Similar to Roy et al. [30], the two "peers" could share the "local use of network service" charge (LuOS) based on the network infrastructure required for the trade to encourage both buyers and sellers to participate in local trading. All consumers can be classified into four categories: (a) without a solar PV system and energy storage, (b) only have a PV system, (c) only have energy storage, (d) with both a solar PV system and an energy storage. In this setting, the consumers can either import energy from the retailer in a business-as-usual (BAU) manner or the P2P market. Similarly, their surplus energy can be exported to either the retailer or P2P market.

In this study, we develop an optimization model for P2P energy trading with a pricing mechanism and distributed energy technologies. The objective function is to minimize the total energy cost by finding the optimal trading decisions and operational decisions related to the solar PV systems and the energy storage for each household in a local community. The rest of this section will describe the proposed mixed integer linear programming (MILP) model for P2P energy trading.

2.1. Input parameters and decision variables

The inputs for the proposed MILP model includes (1) the sets of households and their adopted distributed technologies, demand, and (2) the market and technical signals. Below are the notations and descriptions of input parameters used in this study:

Input parameters:

$\mathscr{A}$: set of households without a PV system and an energy storage

$\mathscr{B}$: set of households with only a PV system installed

$\mathscr{C}$: set of households with only an energy storage installed

$\mathscr{D}$: set of households with a PV system and an energy storage installed

$\mathscr{P} = \{\mathscr{A} \cup \mathscr{B} \cup \mathscr{C} \cup \mathscr{D}\}$: the whole population

$D_{i,t}$: demand (kWh) of household i for each time slot t. Without losing the generality, each time slot here is assumed to be 1 h.

$S_{i,t}$: solar zone indicator which take a binary value (0 or 1) for $i \in \mathscr{B} \cup \mathscr{D}$. If $S_{i,t} = 1$, PV panels can generate energy in time slot t; otherwise, no energy will be generated by the PV panels.

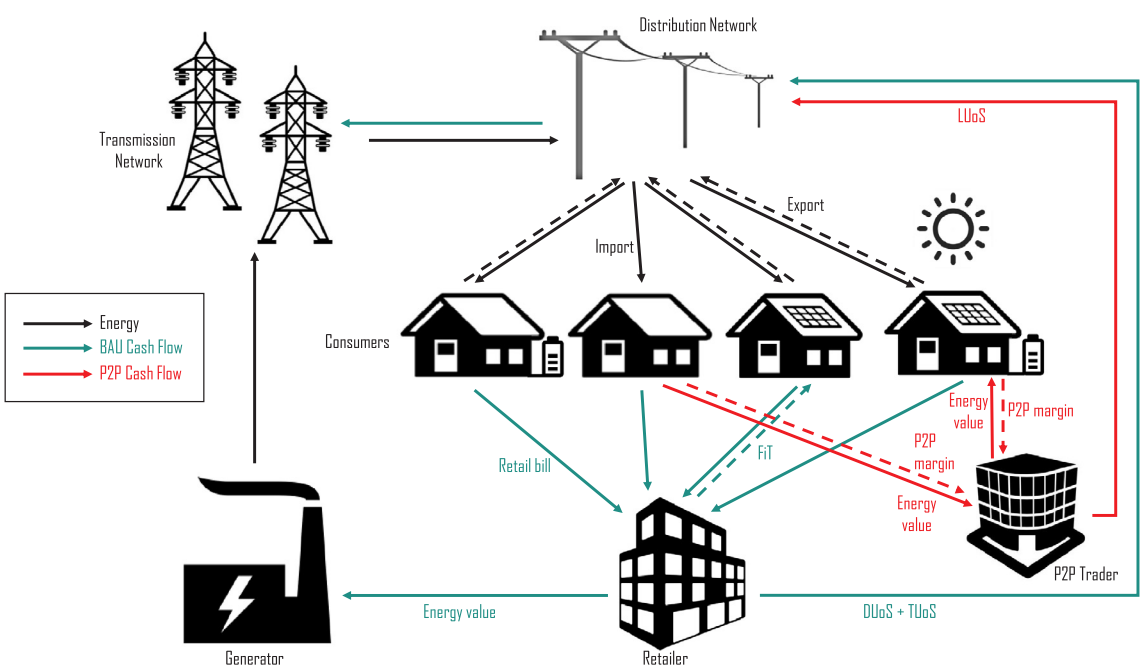

Fig. 1. P2P energy trading model adapted from Roy et al. [30] (BAU: business as usual; LUoS: local use of network service; DUoS: Distribution Use of System; TUoS: Transmission Use of System; FiT: feed-in tariff).

$O_{i,t}$: PV output (kWh) in time t when $S_{i,t} = 1$ for $i \in \mathscr{B} \cup \mathscr{D}$; otherwise, 0.
μ_i: levelized cost of energy for the PV system owned by each household $i \in \mathscr{B} \cup \mathscr{D}$.
$\alpha_{i,t}$: retail price (\$/kWh) in time slot t for each household i.
$\beta_{i,t}$: feed-in tariff (\$/kWh) in time slot t for each household i.
T: number of time slots considered for optimization. It is noted that $t = 1...T$.
λ_i: charging rate (kWh) for $i \in \mathscr{C} \cup \mathscr{D}$.
CMIN_i, CMAX_i: minimum and maximum battery level (kWh) for $i \in \mathscr{C} \cup \mathscr{D}$.
BINIT_i: is the initial battery level at time slot 1.
ω_i: levelized cost of storage for the battery owned by each household $i \in \mathscr{C} \cup \mathscr{D}$.
$\eta_{i,t}$: reservation purchase price (\$/kWh) in time slot t for each household i.
$\gamma_{i,t}$: reservation sale price (\$/kWh) in time slot t for each household i.
θ: P2P trader margin (%) on the total P2P trading amount.
M: is a very large number.

The proposed MILP model will take into account all the above inputs to find the optimal (1) trading decisions such as how much energy generated by PV systems will be traded in the P2P market and (2) operational decisions such as when the energy storage will start charging or discharging. Below are the notations and descriptions of decision variables considered in this study:

Decision variables:
$EGU_{i,t}$: energy purchased from the retailer by household i to use in time t
$EGB_{i,t}$: energy purchased from the retailer to charge the battery of household $i \in \mathscr{C} \cup \mathscr{D}$ in time t
$EPU_{i,t}$: energy purchased from P2P market by household i to use in time t
$EPB_{i,t}$: energy purchased from the P2P market to charge the battery of household $i \in \mathscr{C} \cup \mathscr{D}$ in time t
$ESU_{i,t}$: energy generated by household $i \in \mathscr{B} \cup \mathscr{D}$ to use in time t
$ESB_{i,t}$: energy generated by household $i \in \mathscr{D}$ to charge its battery in time t
$ESG_{i,t}$: energy generated by household $i \in \mathscr{B} \cup \mathscr{D}$ to sell back to the retailer in time t
$ESP_{i,t}$: energy generated by household $i \in \mathscr{B} \cup \mathscr{D}$ to sell in the P2P market in time t
$EBU_{i,t}$: energy discharged from the battery of household $i \in \mathscr{C} \cup \mathscr{D}$ to use in time t
$EBG_{i,t}$: energy discharged from the battery of household $i \in \mathscr{C} \cup \mathscr{D}$ to sell back to the retailer in time t
$EBP_{i,t}$: energy discharged from the battery of household $i \in \mathscr{C} \cup \mathscr{D}$ to sell in the P2P market in time t
$B_{i,t}$: battery level of the battery of household $i \in \mathscr{C} \cup \mathscr{D}$ at the beginning of time slot t
$Y_{i,t}$: charging decision of household $i \in \mathscr{C} \cup \mathscr{D}$ in the time slot t: 1 for charging and 0 otherwise
$Z_{i,t}$: discharging decision of household $i \in \mathscr{C} \cup \mathscr{D}$ in the time slot t: 1 for discharging and 0 otherwise
$I_{i,t}$: P2P purchasing decision of household i in the time slot t: 1 for purchase energy from P2P market and 0 otherwise
$U_{i,t}$: P2P selling decision of household $i \in \mathscr{B} \cup \mathscr{C} \cup \mathscr{D}$ in the time slot t: 1 for selling energy to P2P market and 0 otherwise
P_t: clearing price for the P2P market in the time slot t

2.2. Objective function

$$
\begin{aligned}
&\textbf{\textit{Minimize}}\ \text{Net Total Cost} = \\
&\quad \alpha_{i,t}\left(\sum_{i\in\mathscr{P}}\sum_{t=1}^{T} EGU_{i,t} + \sum_{i\in\mathscr{C}\cup\mathscr{D}}\sum_{t=1}^{T} EGB_{i,t}\right) \\
+ &\quad \eta_{i,t}(1+\theta)\left(\sum_{i\in\mathscr{P}}\sum_{t=1}^{T} EPU_{i,t} + \sum_{i\in\mathscr{C}\cup\mathscr{D}}\sum_{t=1}^{T} EPB_{i,t}\right) \\
+ &\quad \mu_i\left(\sum_{i\in\mathscr{D}}\sum_{t=1}^{T} ESB_{i,t} + \sum_{i\in\mathscr{B}\cup\mathscr{D}}\sum_{t=1}^{T} (ESU_{i,t} + ESG_{i,t} + ESP_{i,t})\right) \\
+ &\quad \omega_i\left(\sum_{i\in\mathscr{D}}\sum_{t=1}^{T} ESB_{i,t} + \sum_{i\in\mathscr{C}\cup\mathscr{D}}\sum_{t=1}^{T} (EGB_{i,t} + EPB_{i,t} + EBG_{i,t} + EBP_{i,t} \right. \\
&\quad \left. + EBG_{i,t})\right) \\
- &\quad \beta_{i,t}\left(\sum_{i\in\mathscr{B}\cup\mathscr{D}}\sum_{t=1}^{T} ESG_{i,t} + \sum_{i\in\mathscr{C}\cup\mathscr{D}}\sum_{t=1}^{T} EBG_{i,t}\right) \\
- &\quad \gamma_{i,t}(1-\theta)\left(\sum_{i\in\mathscr{B}\cup\mathscr{D}}\sum_{t=1}^{T} ESP_{i,t} + \sum_{i\in\mathscr{C}\cup\mathscr{D}}\sum_{t=1}^{T} EBP_{i,t}\right)
\end{aligned}
\tag{1}
$$

The objective function in Eq. (1) is the net energy cost of all households within the planning horizon. The objective function takes into account the cost of purchasing energy from retailers and P2P market, the cost for using PV and energy storage systems, and the revenues earned from selling energy back to retailers and P2P market. The P2P trading margin θ is included to take into account the costs that consumers and prosumers need to pay to the P2P trader for taking care of the transactions and related service costs (e.g., LUoS charge). It is noted the objective value here does not reflect the real net total cost as the energy costs/revenues from the P2P market have to be calculated based on the clearing price P_t rather than the reservation price $\eta_{i,t}$ and $\gamma_{i,t}$. However, replacing the reservation price with the clearing price in the objective function will make the optimization problem non-linear and a lot harder to find the optimal solutions. Moreover, using the clearing price may make the optimal solutions biased toward extreme decisions to favor a small group of households (because the optimal solution will only make trading decisions that help to minimize the net energy cost rather than trying to achieve fair savings for all households). These extreme decisions then discourage people from participating in the P2P market (this long-term behavior is not investigated in this study). Therefore, the objective function (1) is employed in this study and the corrected net total cost (CNTC) will be calculated after the optimal solutions are found to reflect the real cost and benefit of P2P trading.

2.3. Demand constraints

$$
\begin{aligned}
D_{i,t} &= EGU_{i,t} + EPU_{i,t} \quad \forall\, i \in \mathscr{A} \quad \forall\, t = 1...T \\
D_{i,t} &= EGU_{i,t} + ESU_{i,t} + EPU_{i,t} \quad \forall\, i \in \mathscr{B} \quad \forall\, t = 1...T \\
D_{i,t} &= EGU_{i,t} + EBU_{i,t} + EPU_{i,t} \quad \forall\, i \in \mathscr{C} \quad \forall\, t = 1...T \\
D_{i,t} &= EGU_{i,t} + EBU_{i,t} + ESU_{i,t} + EPU_{i,t} \quad \forall\, i \in \mathscr{D} \quad \forall\, t = 1...T
\end{aligned}
\tag{2}
$$

Constraints (2) ensure that the consumer demand will be satisfied. Several household DG portfolios are considered in our model. Therefore supply for demand can be met by several resources. Households belonging to group $\mathscr{A}$ can only purchase energy from their retailers or the P2P market to satisfy the demand. The energy from solar PV systems (groups $\mathscr{B}$ and $\mathscr{D}$) and the batteries (groups $\mathscr{C}$ and $\mathscr{D}$) are only applied if the respective technologies are installed.

Table 1
Battery specification used in the simulation model.

Specification	Powerwall 1	Powerwall 2
Price (AUD)	5500	10300
Capacity (kWh)	7	14
Efficiency (%)	87	89
Maximum discharge rate (kWh)	3.3	5
Cycles (10 years)	7300	7300
LCOS (AUD/kWh)	0.1237	0.1132

2.4. PV output constraints

$$ESU_{i,t} + ESG_{i,t} + ESP_{i,t} \leqslant O_{i,t} \times S_{i,t} \quad \forall\, i \in \mathscr{B} \quad \forall\, t = 1...T$$
$$ESU_{i,t} + ESB_{i,t} + ESG_{i,t} + ESP_{i,t} \leqslant O_{i,t} \times S_{i,t} \quad \forall\, i \in \mathscr{D} \quad \forall\, t = 1...T \tag{3}$$

Constraints (3) ensure that the usage of energy generated by the solar PV systems will not exceed their outputs. The solar energy in our model can be either used for the household loads, charging the battery, selling back to the retailers or the P2P market.

2.5. Battery constraints

$$B_{i,t+1} = B_{i,t} + (EGB_{i,t} + EPB_{i,t}) - (EBG_{i,t} + EBP_{i,t} + EBU_{i,t}) \quad \forall\, i \in \mathscr{C} \quad \forall\, t = 1...T-1$$

$$B_{i,t+1} = B_{i,t} + (EGB_{i,t} + ESB_{i,t} + EPB_{i,t}) - (EBG_{i,t} + EBP_{i,t} + EBU_{i,t}) \quad \forall\, i \in \mathscr{D} \quad \forall\, t = 1...T-1 \tag{4}$$

$$B_{i,1} = \mathrm{BINIT}_i \quad \forall\, i \in \mathscr{C} \cup \mathscr{D} \tag{5}$$

$$\mathrm{CMIN}_i \leqslant B_{i,t} \leqslant \mathrm{CMAX}_i \quad \forall\, i \in \mathscr{C} \cup \mathscr{D} \quad \forall\, t = 1...T \tag{6}$$

$$EGB_{i,t} + EPB_{i,t} \leqslant \lambda_i Y_{i,t} \quad \forall\, i \in \mathscr{C} \quad \forall\, t = 1...T$$
$$EGB_{i,t} + ESB_{i,t} + EPB_{i,t} \leqslant \lambda_i Y_{i,t} \quad \forall\, i \in \mathscr{D} \quad \forall\, t = 1...T \tag{7}$$

$$EBG_{i,t} + EBP_{i,t} + EBU_{i,t} \leqslant \lambda_i Z_{i,t} \quad \forall\, i \in \mathscr{C} \cup \mathscr{D} \quad \forall\, t = 1...T \tag{8}$$

$$Y_{i,t} + Z_{i,t} \leqslant 1 \quad \forall\, i \in \mathscr{C} \cup \mathscr{D} \quad \forall\, t = 1...T \tag{9}$$

Constraints (4)–(9) are used to capture the dynamics of battery level through different time slots. Depending on the charging or discharging decisions in each time slot, the battery level will increase, decrease, or stand still. These constraints reflect the battery states and technological constraints regarding the battery capacity and charging/discharging operations. For each time slot, the battery can only charge or discharge (not both) as shown in constraints (9). These constraints help to reduce the charging/discharging frequency which may deteriorate the battery lifetime. Without constraints (9), the fluctuations of charging and discharging can increase and reduce the lifetime of the battery [31], just to make a minor decrease in the net energy cost. As this study focuses on P2P trading, only a basic battery model is considered. However, more sophisticated and specific battery models in the literature can be employed to fulfill real-world battery constraints on temperature rise and charge time [32] and to develop a state-monitored predictive battery

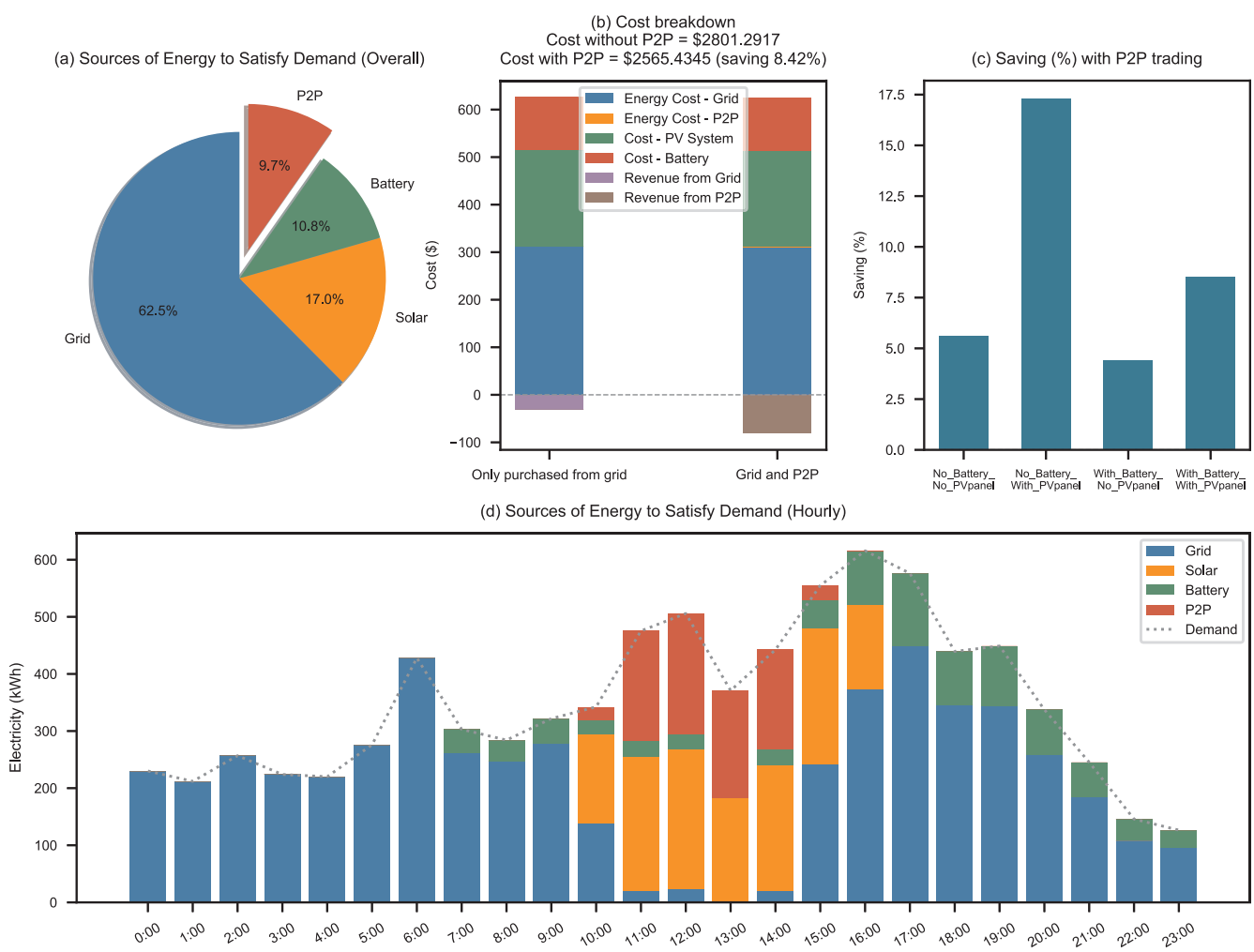

Fig. 2. Electricity consumption patterns on Monday with $\theta = 5\%$, PV area = 8 m^2.

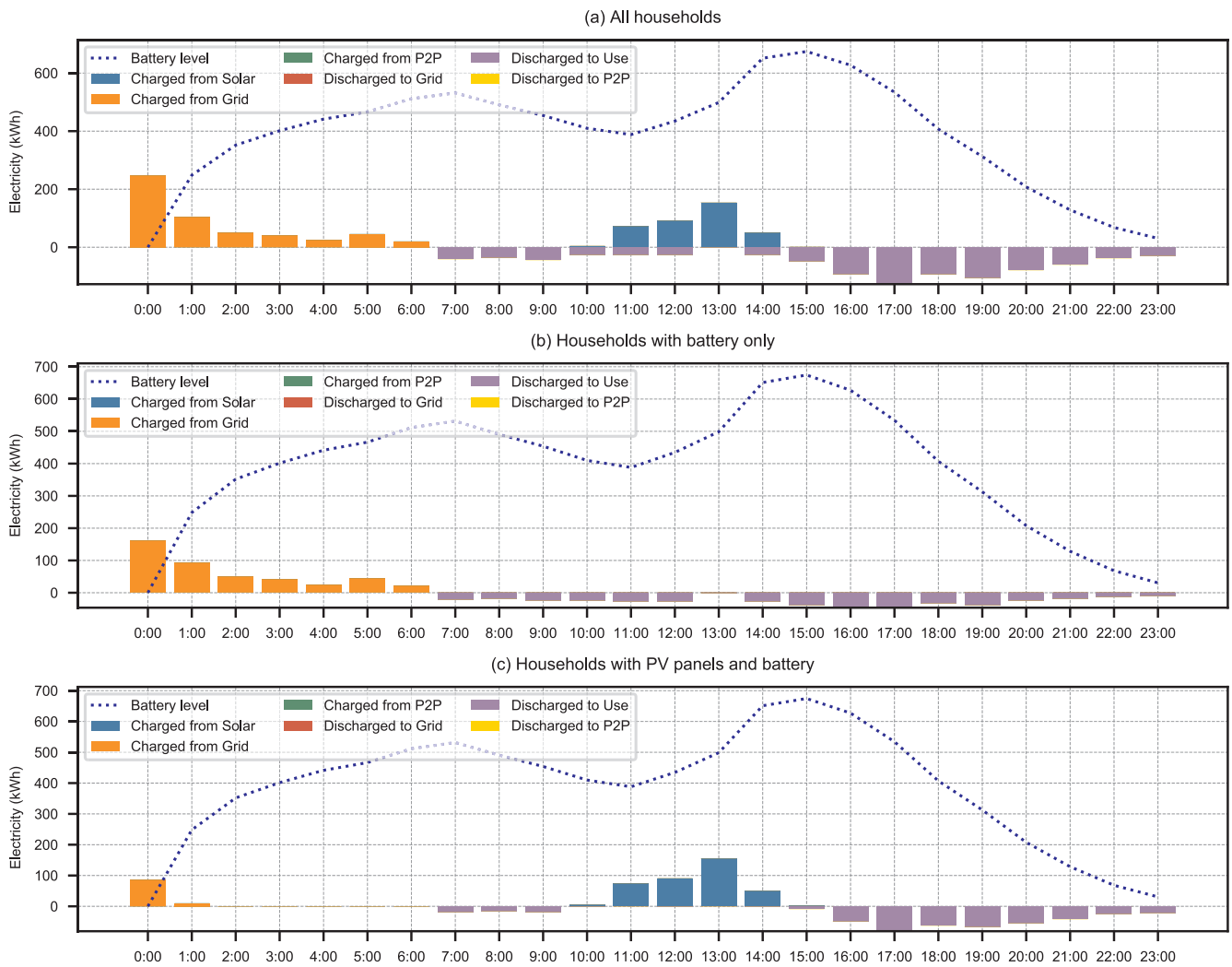

Fig. 3. Battery operations on Monday with $\theta = 5\%$, PV area = 8 m^2.

control system [33]. A highly accurate estimation of battery state-of-charge and state-of-health can be achieved using a multi-time-scale nonlinear algorithm [34] or a Gaussian model-based estimation method [35].

2.6. Trading and pricing constraints

$$\eta_{i,t} + M(1-I_{i,t}) \geqslant P_t \quad \forall\, i \in \mathscr{P} \quad \forall\, t = 1...T \tag{10}$$

$$\gamma_{i,t} - M(1-U_{i,t}) \leqslant P_t \quad \forall\, i \in \mathscr{P} \quad \forall\, t = 1...T \tag{11}$$

$$\begin{aligned} &EPU_{i,t} \leqslant D_{i,t}U_{i,t} \quad \forall\, i \in \mathscr{A} \cup \mathscr{B} \quad \forall\, t = 1...T \\ &EPU_{i,t} + EPB_{i,t} \leqslant D_{i,t}U_{i,t} \quad \forall\, i \in \mathscr{C} \cup \mathscr{D} \quad \forall\, t = 1...T \end{aligned} \tag{12}$$

$$\begin{aligned} &ESP_{i,t} \leqslant (O_{i,t} \times S_{i,t})U_{i,t} \quad \forall\, i \in \mathscr{B} \quad \forall\, t = 1...T \\ &EBP_{i,t} \leqslant \mathrm{CMAX}_i U_{i,t} \quad \forall\, i \in \mathscr{C} \quad \forall\, t = 1...T \\ &ESP_{i,t} + EBP_{i,t} \leqslant (\mathrm{CMAX}_i + O_{i,t} \times S_{i,t})U_{i,t} \quad \forall\, i \in \mathscr{D} \quad \forall\, t = 1...T \end{aligned} \tag{13}$$

$$I_{i,t} + U_{i,t} \leqslant 1 \quad \forall\, i \in \mathscr{P} \quad \forall\, t = 1...T \tag{14}$$

$$\sum_{i \in \mathscr{P}} EPU_{i,t} + \sum_{i \in \mathscr{C} \cup \mathscr{D}} EPB_{i,t} = \sum_{i \in \mathscr{B} \cup \mathscr{D}} ESP_{i,t} + \sum_{i \in \mathscr{C} \cup \mathscr{D}} EBP_{i,t} \quad \forall\, t = 1...T \tag{15}$$

Constraints (10)–(15) model the pricing mechanism used by the P2P market in each time slot t. Constraints (10) ensure that a household can only purchase energy from the P2P market at time slot t if the reservation purchase price is higher than the clearing price P_t. Similarly, constraints (11) ensure that a household can only sell energy to the P2P market at time slot i if their reservation sale price is lower than the clearing price P_t. Constraints (12) and (13) shows which resources are used to export energy to the P2P market and how the energy purchased from the P2P market is allocated. Constraints (14) restricted a household from being both buyer and seller in the P2P market in a particular time slot. Constraints (15) ensures that the trading amount in the P2P market is balanced.

In addition, all decision variables need to satisfy constraints (16):

$$\begin{aligned} &EGU_{i,t}, EGB_{i,t}, EPU_{i,t}, EPB_{i,t}, ESU_{i,t}, ESB_{i,t}, ESG_{i,t}, ESP_{i,t}, \\ &EBU_{i,t}, EBG_{i,t}, EBP_{i,t}, B_{i,t}, P_t \geqslant 0 \quad \forall\, i, t \\ &Y_{i,t}, Z_{i,t}, I_{i,t}, U_{i,t} \quad \text{are binary} \end{aligned} \tag{16}$$

To solve this MILP model, we use Gurobi 7.5, the state-of-the-art commercial MILP solvers [36,37], on the Dual Intel Xeon Processor E5-2697 with 128 GB Ram. For all experiments in this study, Gurobi will terminate after 30 min or when the relative optimality gap is lower than 0.5%.

3. Case studies

To illustrate the optimization model, we simulate a local community with 500 households based on the real-world data. The customer load shapes examined in this paper is obtained by Supervisory Control and

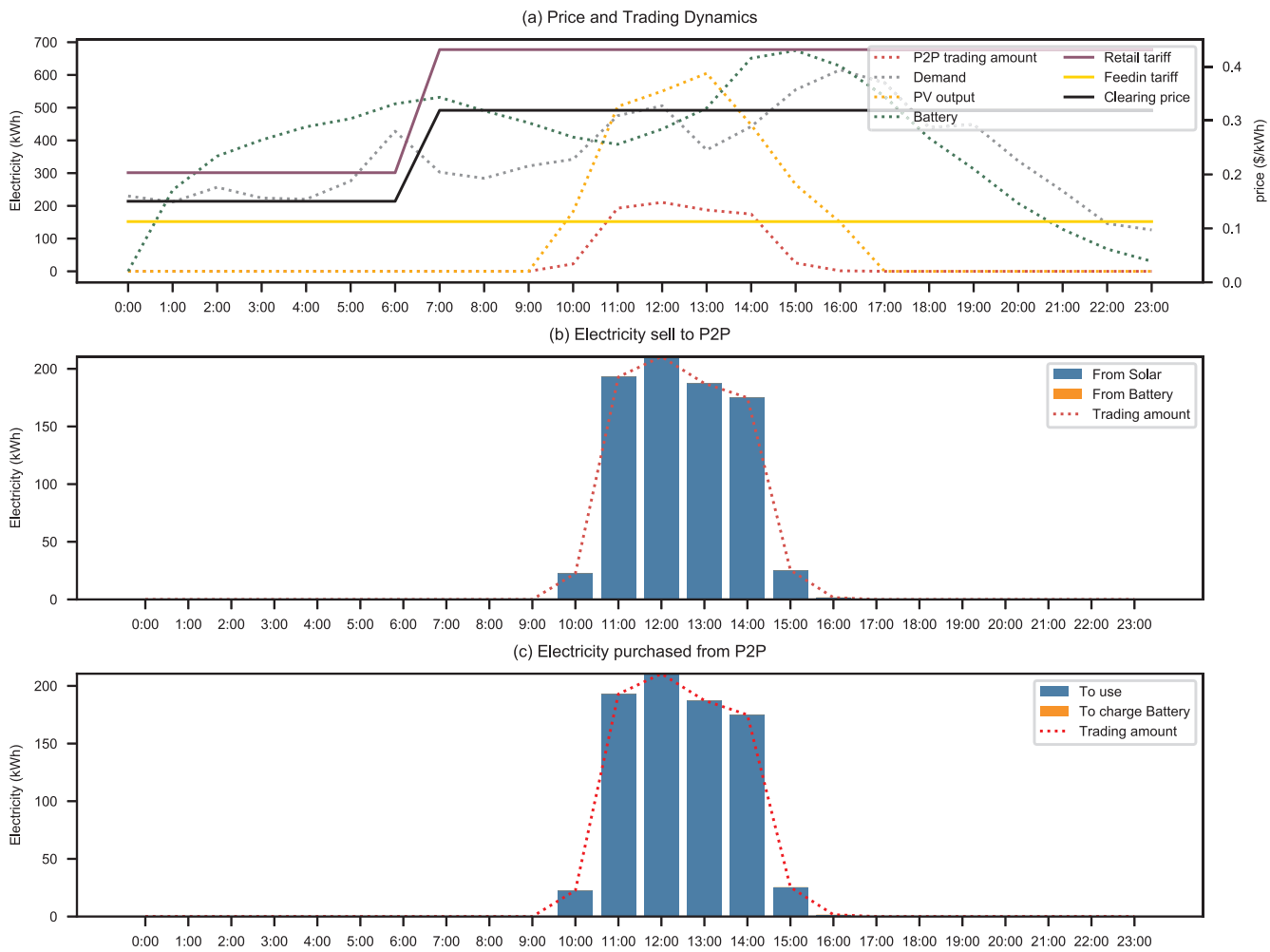

Fig. 4. P2P trading activities on Monday with $\theta = 5\%$, PV area = 8 m^2. Graphs (b) and (c) in this case are identical because batteries are not used, and only energy generated by PV systems are traded in the P2P market.

Data Acquisition (SCADA) and smart meter measurements over a summer week from 15 (Sunday) to 21 February 2015 (Saturday) from eight customers [38]. This data set is selected to test the proposed optimization model because it reflects a wide range of consumer behavior. To further enhance the diversity of consumer load profiles, we randomly scale the load shapes based on the average electricity use by households of different sizes (from 3 to 5 people, i.e., from 12.06 kWh to 21.07 kWh) in Victoria, Australia [39]. The solar profile representing a unit of solar radiation PV captured (watt per square meter) was provided by a third party collaborator based on data captured from the sensor web laboratory in Tasmania [8]. Also, we add some randomness to the solar profiles with a simple random number generator with a normal distribution and a mean and standard deviation of 0 and 0.1 respectively.

The electricity retail price is based on the 2018 "Residential 5-Day Time of Use" plan from Origin, a major power supplier in Australia [40]. Within this plan, the off-peak rate of AUD0.43186/kWh is applied for weekdays, from 7am-11 pm. In other times, an off-peak rate of AUD0.20350/kWh is applied. The daily supply charge is not considered in our simulation as we assume that all households have to pay this amount regardless of their behavior. The feed-in tariff used in this study is AUD0.113/kWh based on the minimum feed-in tariff of Victoria [41]. Similar to the solar profiles, we also add some randomness to the retail price based on a normal distribution with a mean and standard deviation of 0 and 0.05 respectively.

The PV systems examined in our simulation is based on the specifications used in [8]. The PV panel size range from 8 m^2 to 32 m^2 with an efficiency of 13%. In this study, we apply the levelized cost of energy (LCOE) reported by the Australian Energy Council for medium and large PV systems (from 3 kWh to 10 kWh) [42]. LCOE calculated based on the ten-year average mortgage rate for Melbourne is from AUD0.15/kWh to AUD0.16/kWh. For simplicity, we only use LOCE = AUD0.15/kWh in our simulation.

For the battery, we use the specification from the Powerwall 1 and Powerwall 2 from Tesla [43,44], two popular energy storage solutions in the market. The details of these two batteries are provided in Table 1. The price in this table is the approximate installation cost. With P2P trading and the availability of PV systems, we assume that the battery will be used regularly at the frequency of two cycles per day to match with the system design in which the battery can be charged from either the solar PV or the grid (i.e., tariff arbitrage). The levelized cost of storage (LCOS) is determined as:

$$\text{LCOS} = \frac{\text{Price}}{\text{Capacity}\times\text{Cycles}\times\text{Efficiency}} \tag{17}$$

As mentioned in the previous section (at Section 2.3), there are four groups of households investigated in this study categorized based on their adopted technologies. In our simulation, we will set up a theoretical situation in which 50% of the households have PV systems and 50% of them will have energy storage (randomly picked from the two models in Table 1). With this setting, the simulated households will be equally distributed into the four mentioned groups.

The reservation purchase/sale price is the point at which a

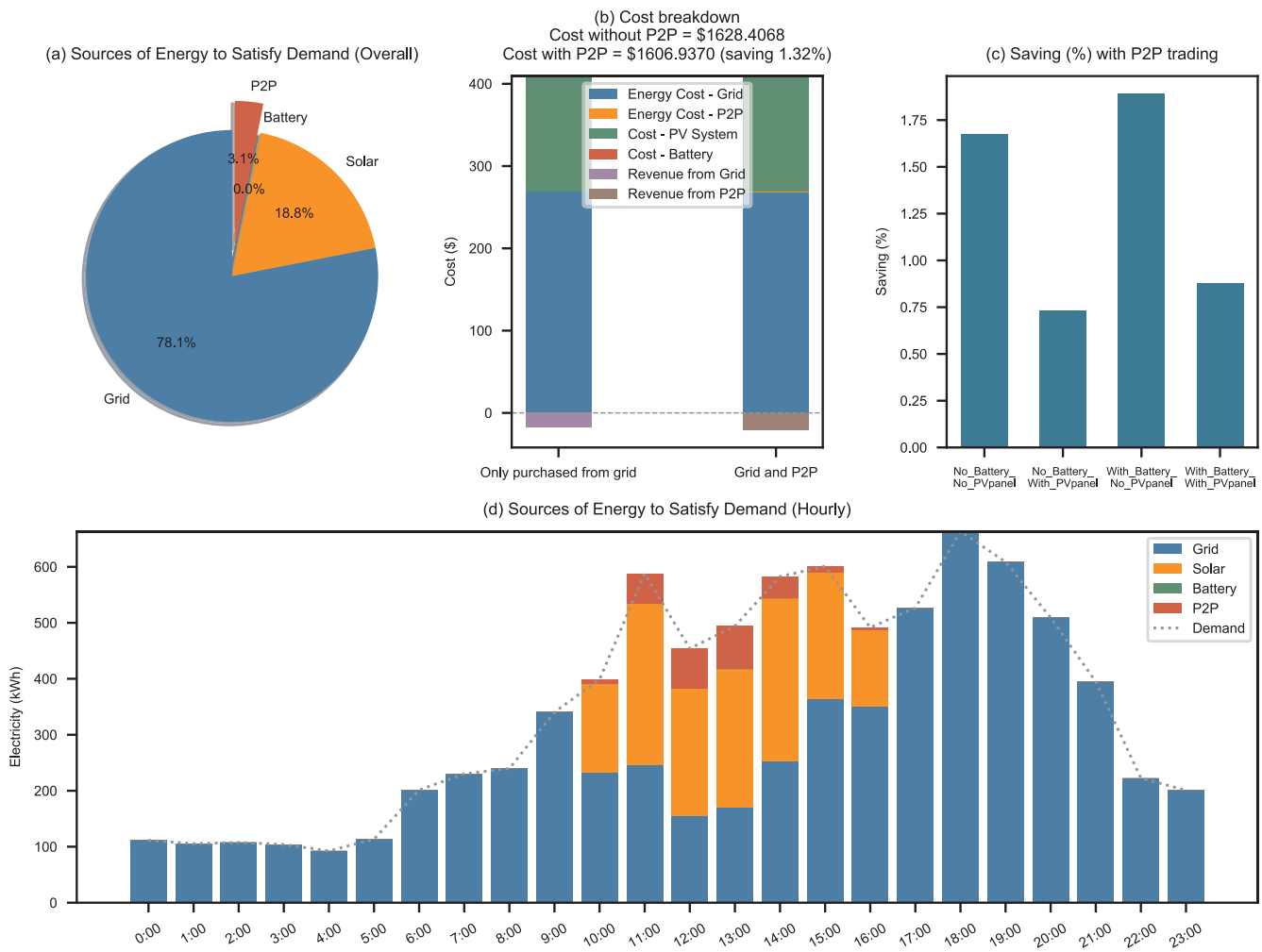

Fig. 5. Electricity consumption patterns on Sunday with $\theta = 5\%$, PV area $= 8\,\mathrm{m}^2$.

consumer or a prosumer is willing to trade on the P2P market. Roy et al. [30] calculate these prices by taking into account the P2P trader margin and LuOS. In this study, we ignore the LuOS charge and assume that this charge is to be covered by the P2P trader margin. We also add a desired margin to the reservation price to represent the consumer/prosumer motivations to participate in P2P trading. In summary, the reservation purchase price (RPP) and the reservation sale price (RSP) can be calculated as:

$$\mathrm{RPP} = \frac{1-\text{Desired Margin}}{1+\text{P2P Trader Margin}} \times \text{Retail Price} \tag{18}$$

$$\mathrm{RSP} = \frac{1+\text{Desired Margin}}{1-\text{P2P Trader Margin}} \times \text{Feed-in Tariff} \tag{19}$$

Based on this calculation, it is easy to see that the higher desired margin and the higher P2P trader margin (i.e., θ in the optimization model) will make the reservation purchase price lower and the reservation sale price higher. When the gaps between these two reservation prices are more substantial, the consumers/prosumers will be less likely to participate in trading in the P2P market. In our simulation, we set the desired margin of 10% for all households, and different P2P trader margin values are investigated ranging from 5% to 20%.

4. Results and discussion

This section illustrates the results obtained with the proposed optimization model. We will use the results obtained from Sunday and Monday to illustrate the optimal decisions found by the optimization model. Then, we compare how the energy costs can be saved in different simulation scenarios. Finally, the savings related to each group of households are reported.

4.1. Optimal trading and operational patterns

Fig. 2 illustrates the aggregate energy consumption patterns on Monday given that the P2P margin is 5% and the PV area is $8\mathrm{m}^2$ (for all households with PV systems installed). The pie chart in Fig. 2(a) shows the energy sources consumed by households on the corresponding day. As only 50% of the households in our simulation possess PV systems, it is not surprising that the primary energy source is from the grid. The second largest source of energy is from PV systems, followed by battery and P2P market. The energy sources are broken down for each hour in Fig. 2(d). We can see that households need to import electricity from the grid in the early morning to charge the battery as the (current) retail price is low. In the day, around 10:00, PV systems start to generate electricity to use and export to the grid (i.e., back to the retailer and so non-participating customers) or the P2P market. From 10:00 to 16:00, the PV outputs can satisfy roughly 50% of the electricity demand in the simulated community. Around noon time, as the PV outputs exceed the demand of the prosumers, the surplus electricity is traded in the P2P market and charge the battery.

Details about the battery operations and P2P trading activities are shown in Figs. 3 and 4 respectively. More specifically, Fig. 3 shows how households ($\mathscr{C}$ and $\mathscr{D}$) charge and discharge their batteries. For

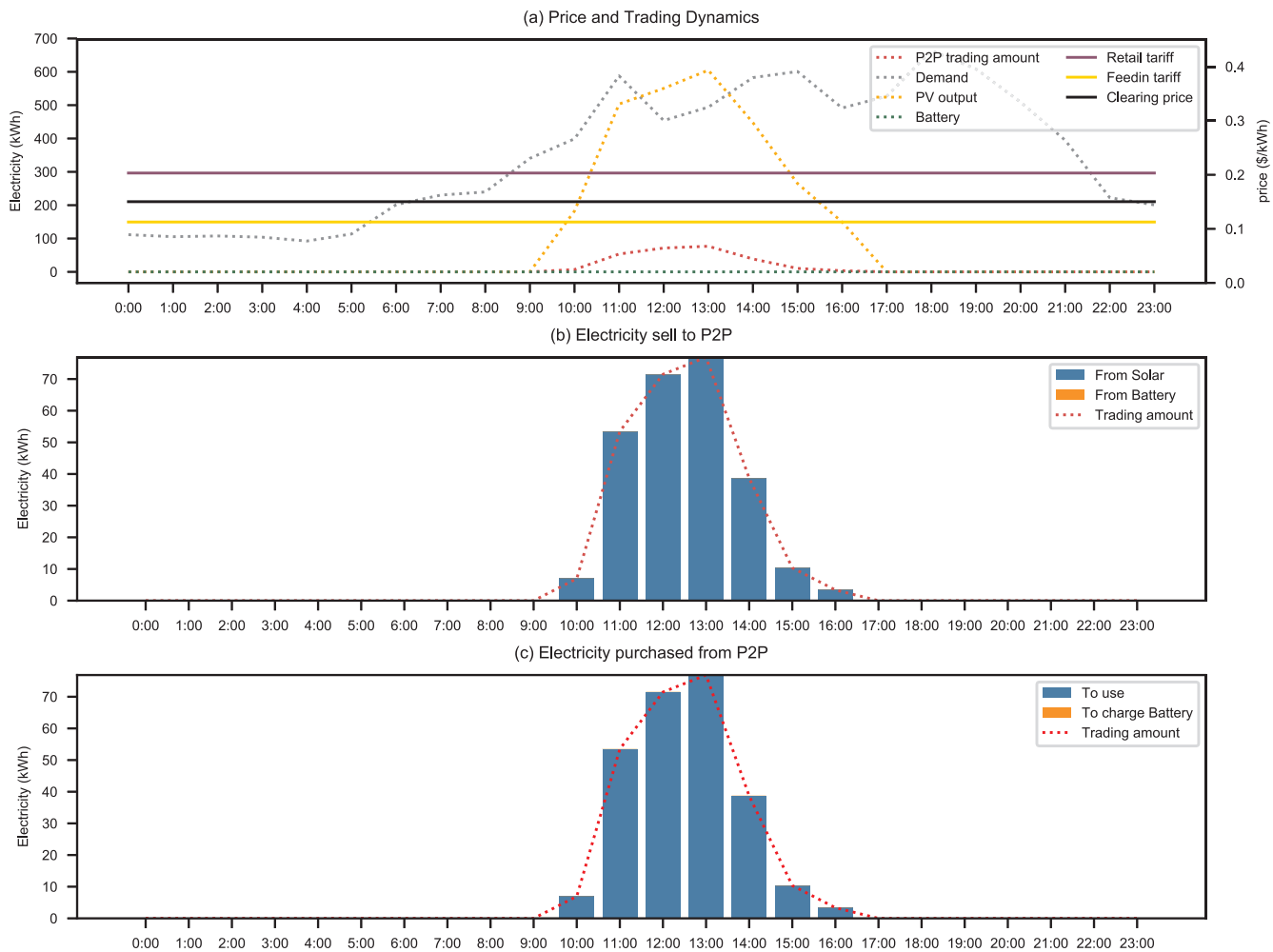

Fig. 6. P2P trading activities on Sunday with $\theta = 5\%$, PV area $= 8\,\mathrm{m}^2$. Graphs (b) and (c) in this case are identical because batteries are not used, and only energy generated by PV systems are traded in the P2P market.

households in group $\mathscr{C}$, their batteries are charged from the grid from 0:00 to 6:00 when the retail tariff is low (off-peak) and discharged during peak hours. For households in group $\mathscr{D}$, the batteries are charged twice per day, once with PV systems and once with the grid. When the PV outputs are not available, the batteries are discharged to provide electricity to households. From Fig. 2(b), savings are mainly achieved through higher revenue from exporting electricity to the P2P market. Because the electricity sold in the P2P market is only from PV systems, households with PV systems ($\mathscr{B}$ and $\mathscr{D}$) achieve the most favorable savings. Fig. 4(a) shows the clearing price for P2P trading throughout the day. It is easy to see that households will save more during peak hours. Fig. 4(b) and (c) show which energy sources are used to export energy to the grid and when energy purchased from the P2P market is used. In this experiment, only surplus energy from PV systems is exported to P2P market as households that purchased energy from the P2P market only use it to satisfy their demands directly. Batteries do not play a critical role in P2P energy trading since the operational cost is still quite high.

The optimal solution for Sunday is shown in Figs. 5 and 6. Owing to the retail price being low on Sunday, it is less attractive to trade surplus electricity in the P2P market (overall saving is only 1.32%). Only a small amount of electricity is traded from 10:00 to 15:00 as the prosumers cannot self-consume all electricity generated by PV systems. Note, battery storage is not used at all in this scenario because LCOS is too high as compared to the retail price.

4.2. Sensitivity analyses

In this section, we will further examine the sensitivity of input parameters on the optimal solutions. The first part will focus on the scale of PV systems and P2P margin. The second part emphasizes two key market signals, i.e., feed-in tariff, and PV penetration.

4.2.1. The impact of PV systems and P2P margins on P2P trading

Fig. 7 shows the average savings (for all households) by participating in P2P trading given different combinations of PV systems and P2P margins. Across the seven simulated days, we can see a distinct pattern, i.e., households will save more money when the P2P margin decreases and larger PV systems are installed. This pattern is not surprising as a low P2P margin will make it easier to find a clearing price that satisfies many participants and encouraging P2P trading. On the other hand, larger PV systems will produce more affordable electricity to satisfy the demand of households in the community. As discussed earlier, the savings are lowest on the weekend as the retail price is low. There is roughly no saving during the weekend if the P2P margin is 20%. The maximal saving (16%) is achieved on Friday if the P2P margin is 5% and PV area $= 32\,\mathrm{m}^2$.

In general, the savings are more sensitive to the scale of PV systems installed in the community than the P2P margin. For example, in Fig. 7(f), the saving decreases by only 5% (from 16% to 11%) when the P2P margin is four times larger (5–20%). However, the saving drops by half (from 16% to 8.2%) when the PV area decreases by fourfold from

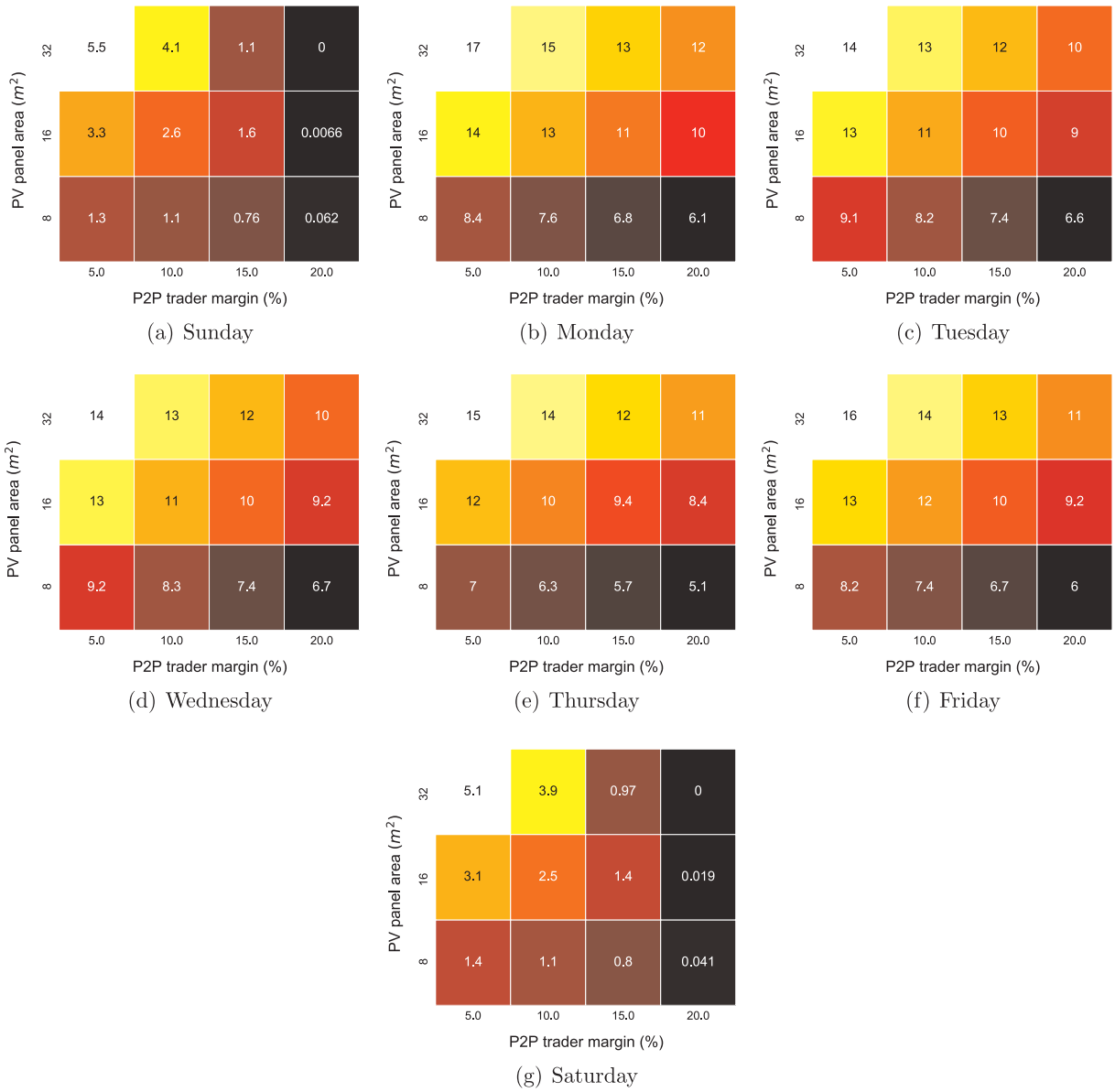

Fig. 7. Saving from participating in the P2P energy trading – PV area vs P2P margin.

32 m^2 to 8 m^2. It suggests that the economic advantages of P2P trading will strongly depend on the PV penetration of the local community.

The savings for each household DG ($\mathscr{A}$, $\mathscr{B}$, $\mathscr{C}$, $\mathscr{D}$) are presented in Table 2. Some interesting patterns are:

- *Households with PV systems ($\mathscr{B}$ and $\mathscr{D}$) have the highest savings, especially during the weekdays*: this is consistent with our early results as most owners of PV systems can export their surplus electricity at a higher price via P2P market as compared to the feed-in tariff.
- *Households achieve maximal savings with both PV systems and batteries installed ($\mathscr{D}$), but only when the PV systems are large*: prosumers with small PV systems will be more likely to immediately consume all electricity generated by their PV systems either to satisfy their power demand or to export to the P2P market (if their demand is low). Therefore, there is only a small amount of cheap and clean electricity generated from small PV systems to be stored. The prosumers can only take full advantages of batteries when their PV systems are large enough. This phenomenon is better visualized in Fig. 8.
- *When small or medium scale PV systems are adopted, households with only PV systems installed ($\mathscr{B}$) are more beneficial*: the main reason is that these households do not consider LCOS in their total costs because they do not install any battery storage.
- *Households with batteries ($\mathscr{C}$) only do not show any advantages and they actually have slightly lower savings as compared to households without adopting any technologies*: the households with batteries only can only benefit from purchasing cheap electricity through the P2P market but have to take into account LCOS for using their batteries (which is energy for the households without adopting any technologies).
- *Saving of households without PV systems ($\mathscr{A}$ and $\mathscr{C}$) are not sensitive to P2P margin*: these households are only sensitive to the number of PV outputs in the community, governed by the size of the PV systems. The main reason is that the amount of electricity each of these households purchases from the P2P market is relatively low as compared to the amount of electricity exported by owners of PV systems.

In general, the results from the optimization model show that households can reduce their energy costs by participating in P2P

Table 2
Saving (%) by participating in P2P trading – PV area vs P2P margin.

PV area	P2P margin (θ)	Sun.	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	*Average*
				Without PV Systems and Battery ($\mathscr{A}$)					
8	5.0	1.68	5.63	5.86	5.99	4.60	5.50	1.74	*4.43*
	10.0	1.26	5.63	5.86	5.99	4.60	5.50	1.29	*4.30*
	15.0	0.81	5.64	5.86	5.99	4.60	5.50	0.83	*4.17*
	20.0	0.08	5.63	5.86	5.99	4.60	5.50	0.05	*3.96*
16	5.0	4.04	8.72	7.82	7.81	7.35	8.04	3.87	*6.81*
	10.0	2.92	8.72	7.82	7.81	7.35	8.04	2.78	*6.49*
	15.0	1.75	8.72	7.82	7.81	7.35	8.04	1.62	*6.16*
	20.0	0.01	8.72	7.82	7.81	7.35	8.04	0.02	*5.68*
32	5.0	6.97	10.00	8.61	8.66	9.27	9.46	6.51	*8.50*
	10.0	4.89	10.00	8.61	8.66	9.27	9.46	4.64	*7.93*
	15.0	1.31	10.00	8.61	8.66	9.27	9.46	1.12	*6.92*
	20.0	0.00	10.00	8.61	8.66	9.27	9.46	0.00	*6.57*
				With PV Systems Only ($\mathscr{B}$)					
8	5.0	0.73	17.32	18.89	18.93	13.41	17.09	0.73	*12.44*
	10.0	0.68	14.69	16.04	16.11	11.36	14.49	0.69	*10.58*
	15.0	0.60	12.29	13.43	13.55	9.48	12.12	0.60	*8.87*
	20.0	0.06	10.08	11.05	11.19	7.77	9.94	0.04	*7.16*
16	5.0	1.73	22.64	20.11	20.23	17.54	20.47	1.66	*14.91*
	10.0	1.62	19.13	16.97	17.07	14.82	17.28	1.56	*12.63*
	15.0	1.04	15.92	14.10	14.19	12.33	14.36	0.80	*10.39*
	20.0	0.01	12.99	11.47	11.54	10.05	11.69	0.02	*8.25*
32	5.0	2.97	23.74	20.41	19.74	22.61	22.27	2.80	*16.36*
	10.0	2.55	19.99	17.17	16.60	19.04	18.74	2.33	*13.77*
	15.0	0.67	16.58	14.21	13.74	15.78	15.52	0.55	*11.01*
	20.0	0.00	13.44	11.49	11.12	12.79	12.57	0.00	*8.77*
				With Battery Only ($\mathscr{C}$)					
8	5.0	1.89	4.39	5.03	4.72	4.05	4.51	2.09	*3.81*
	10.0	1.42	4.39	5.03	4.72	4.05	4.51	1.55	*3.67*
	15.0	0.88	4.39	5.03	4.72	4.05	4.51	0.98	*3.51*
	20.0	0.05	4.39	5.03	4.72	4.05	4.51	0.03	*3.26*
16	5.0	4.74	7.90	7.41	7.58	6.68	7.40	4.63	*6.62*
	10.0	3.43	7.90	7.41	7.58	6.68	7.40	3.33	*6.25*
	15.0	2.13	7.90	7.41	7.58	6.68	7.40	2.00	*5.87*
	20.0	0.00	7.90	7.41	7.58	6.68	7.40	0.02	*5.28*
32	5.0	7.74	9.67	8.88	8.78	9.09	9.43	7.17	*8.68*
	10.0	5.50	9.67	8.88	8.78	9.09	9.43	5.19	*8.08*
	15.0	1.52	9.67	8.88	8.78	9.09	9.43	1.43	*6.97*
	20.0	0.00	9.67	8.88	8.78	9.09	9.43	0.00	*6.55*
				With PV Systems and Battery ($\mathscr{D}$)					
8	5.0	0.88	8.51	8.90	9.09	7.38	7.69	0.94	*6.20*
	10.0	0.81	7.21	7.55	7.75	6.25	6.51	0.88	*5.28*
	15.0	0.72	6.03	6.31	6.52	5.23	5.43	0.77	*4.43*
	20.0	0.06	4.95	5.17	5.40	4.29	4.44	0.03	*3.48*
16	5.0	2.27	21.32	18.25	18.56	18.02	19.14	2.19	*14.25*
	10.0	2.13	18.02	15.40	15.66	15.23	16.16	2.06	*12.09*
	15.0	1.41	15.01	12.80	13.02	12.68	13.44	1.22	*9.94*
	20.0	0.00	12.24	10.42	10.60	10.35	10.94	0.02	*7.80*
32	5.0	3.86	28.12	23.38	23.95	24.29	26.26	3.53	*19.06*
	10.0	3.43	23.69	19.67	20.15	20.46	22.11	3.08	*16.09*
	15.0	0.90	19.65	16.29	16.69	16.97	18.32	0.70	*12.79*
	20.0	0.00	15.94	13.19	13.51	13.76	14.84	0.00	*10.18*

trading. Depending on their adopted DG technologies, the savings can vary significantly. Given the theoretical scenarios (50% households have PV panels, and 50% households have battery storage), P2P trading is a very attractive option, compared to the feed-in tariff, for the owners of PV systems. The households with only PV systems can save from 7.16% to 14.91% on the energy costs per week. The households with both technologies can save from 3.48% to 14.25% on the energy cost week with small and medium PV systems. If this group installs larger PV systems, their savings can significantly increase (up to 19.06%). The maximal savings can be achieved by prosumers equipped with larger

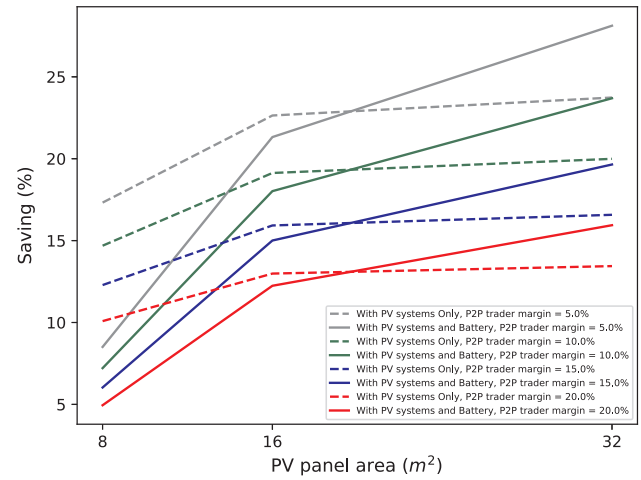

Fig. 8. Saving (%) of groups $\mathscr{B}$ and $\mathscr{D}$ – PV area vs P2P margin.

PV systems and battery storage during weekdays (up to 28.12% as shown in Table 2). Households without any DG technology can also benefit from P2P trading with their savings ranging from 3.96% to 8.5% per week. Households with only battery storage are no better than households without any DG technology regarding saving (from 3.26% to 8.68%) as the costs of battery storage (i.e., LCOS) are still very high. Overall, P2P trading investigated in this study is a feasible and attractive approach to optimize the benefits and values of DG technologies.

4.2.2. The impact of feed-in tariffs and PV penetration

Fig. 9 shows the average savings (for all households) by participating in P2P trading given different combinations of feed-in tariffs and PV penetration (% of households with installed PV systems). In these simulations, we use $\theta = 5\%$, PV area $= 8\,\mathrm{m}^2$. In contrast to the previous section, the impacts of feed-in tariffs and PV penetration are more complex. In our experiments, household savings do not naturally increase when the PV penetration increases or when the feed-in tariff decreases. During weekdays (except for Thursday), the highest savings are achieved when the PV penetration is 50% or 60%. These observations imply that there is an optimal PV penetration at which the maximum saving is obtained. Regarding the feed-in tariff, the savings are not very sensitive during the weekdays. Similar to the previous section, on Saturday and Sunday, there is no impressive saving and the highest saving is achieved when the feed-in tariff is AUD0.13/kWh.

The savings for each household ($\mathscr{A}$, $\mathscr{B}$, $\mathscr{C}$, $\mathscr{D}$) are presented in Table 3. Some interesting patterns are:

- *Households with PV systems ($\mathscr{B}$ and $\mathscr{D}$) have higher savings when the PV penetration is low*: this is reasonable as the higher PV penetration means that there is more surplus energy to be traded on the P2P market, which in turn reduce the clearing price and the saving gains from those who own PV systems.
- *Households without PV systems ($\mathscr{A}$ and $\mathscr{C}$) can have slightly higher savings when the PV penetration increases*: as more surplus energy is available via P2P energy market, it is easier for households without any DG technologies to buy cheap energy to reduce their energy costs. However, it is noted that these households are much less sensitive to the PV penetration than those with PV systems installed.
- *Households with only PV systems installed ($\mathscr{B}$) are more beneficial when the feed-in tariffs are low*: this is easy to understand as the profit obtained from trading the surplus energy on the P2P market will be higher than that obtained by selling energy back to the grid.
- *Households without PV systems installed ($\mathscr{A}$ and $\mathscr{C}$) are not sensitive to the changes of feed-in tariffs*: this is straightforward as these households do not gain any revenue via feed-in tariffs (minor differences are due to small changes in trading and operational decisions).

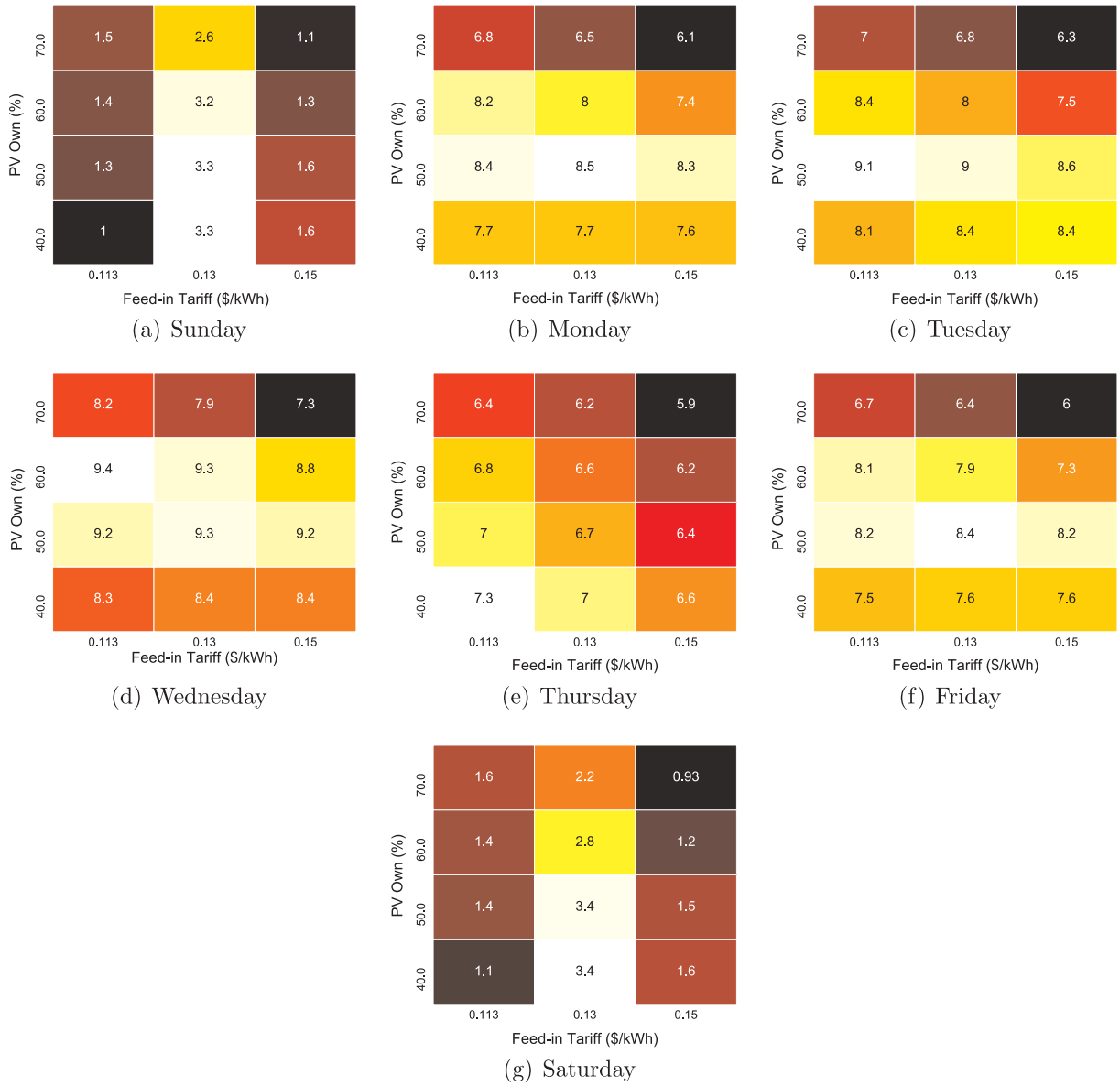

Fig. 9. Saving from participating in the P2P energy trading – Feed-in tariff vs PV penetration.

- *Savings for Households with both PV systems and batteries ($\mathscr{D}$) depend on the combinations of the PV penetration and the feed-in tariff*: from Table 3 and Fig. 10, those households can save more with a higher feed-in tariff when the PV penetration is 40% or 50%. When the PV penetration is more than 50%, these households will save more if the feed-in tariff is low.

The analyses in this section show that savings for households with PV systems installed are highly sensitive to the PV penetration. As the PV penetration increases over time, those who have already owned PV systems will see a rapid drop in their savings. For households which plan to install PV systems, they need to be extra careful and make sure that their installation costs will be worth more than their future savings.

4.3. Further discussion

This study considers a centralized optimization model which tries to minimize the total energy costs of all households. Therefore, the global optimal solutions will not be the optimal solution for each household. It explains why households with both DG technologies, supposedly with higher chances of saving on energy costs, actually do not save as much as households with only PV systems in some scenarios. In the long term, if the households with higher investments in DG technologies think that their contributions to minimizing the total costs to the communities are not suitably equivalent to their savings, they can decide to increase their desired margin. Similarly, households without any DG technology have to decrease their desired margin to be able to purchase electricity from the P2P market. In that case, the savings of different groups of households will change accordingly. We believe that the market will be self-stabilized in the long term under this trading mechanism. This issue will be further investigated in our future studies.

Many countries have a price on carbon emissions as a mechanism to signal for a reduction in emissions and comply to the 2°C global temperature target [45]. This analysis assumes any price on emissions is built into the retail price. However, if a price on carbon is further imposed, utilizing household renewable generation will obviously further improve the cost benefit to the household.

The large-scale deployment of renewable generation and DG with storage poses challenges for the power industry to manage baseload generation in the way that the load and power quality are met with appropriate supply. For example, the electricity consumption pattern shown in Fig. 2 between 10:00 and 14:00 requires minimal or no grid

Table 3
Saving (%) by participating in P2P trading – Feed-in tariff vs PV penetration

PV Own	Feed-in tariff	Sun.	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.	*Average*
				Without PV Systems and Battery ($\mathscr{A}$)					
40.0	0.113	1.18	4.57	4.77	4.88	4.19	4.56	1.15	*3.61*
	0.130	2.78	4.74	5.12	5.18	4.26	4.73	2.83	*4.24*
	0.150	1.39	5.00	5.37	5.41	4.29	4.99	1.29	*3.96*
50.0	0.113	1.68	5.63	5.86	5.99	4.60	5.50	1.74	*4.43*
	0.130	3.30	5.88	5.88	6.24	4.69	5.76	3.66	*5.06*
	0.150	1.55	5.93	5.91	6.41	4.80	5.85	1.50	*4.56*
60.0	0.113	2.06	6.29	6.31	6.88	5.41	6.23	2.09	*5.04*
	0.130	4.18	6.31	6.33	6.95	5.46	6.27	3.86	*5.62*
	0.150	1.68	6.33	6.34	6.96	5.57	6.30	1.33	*4.93*
70.0	0.113	2.57	6.53	6.62	7.09	6.16	6.42	2.60	*5.43*
	0.130	4.15	6.54	6.62	7.10	6.18	6.47	3.65	*5.82*
	0.150	1.37	6.59	6.62	7.12	6.18	6.53	1.18	*5.09*
				With PV Systems Only ($\mathscr{B}$)					
40.0	0.113	0.67	19.49	20.42	20.78	17.86	18.96	0.72	*14.13*
	0.130	3.80	18.18	19.04	19.44	16.56	17.72	4.13	*14.12*
	0.150	2.01	16.36	17.25	17.59	14.86	16.01	1.84	*12.27*
50.0	0.113	0.73	17.32	18.89	18.93	13.41	17.09	0.73	*12.44*
	0.130	3.07	16.24	17.78	17.74	12.37	15.98	2.79	*12.28*
	0.150	1.46	14.53	16.10	16.04	11.08	14.29	1.48	*10.71*
60.0	0.113	0.66	13.45	13.44	15.58	10.19	13.34	0.63	*9.61*
	0.130	2.30	12.70	12.53	14.76	9.35	12.41	1.81	*9.41*
	0.150	0.97	11.30	11.20	13.23	8.29	11.08	0.71	*8.11*
70.0	0.113	0.62	9.00	9.45	11.40	8.08	8.93	0.61	*6.87*
	0.130	1.68	8.30	8.74	10.63	7.44	8.19	1.22	*6.60*
	0.150	0.71	7.30	7.74	9.47	6.57	7.27	0.56	*5.66*
				With Battery Only ($\mathscr{C}$)					
40.0	0.113	1.25	3.57	3.91	3.81	3.58	3.57	1.38	*3.01*
	0.130	3.16	3.84	4.21	4.09	3.70	3.90	3.21	*3.73*
	0.150	1.60	4.09	4.63	4.40	3.76	4.19	1.58	*3.46*
50.0	0.113	1.89	4.39	5.03	4.72	4.05	4.51	2.09	*3.81*
	0.130	3.71	4.68	5.48	5.12	4.11	4.86	4.04	*4.57*
	0.150	1.87	5.13	5.74	5.54	4.15	5.28	1.91	*4.23*
60.0	0.113	2.50	5.44	5.72	5.82	4.65	5.62	2.76	*4.64*
	0.130	4.56	5.82	5.93	6.29	4.76	5.91	4.35	*5.37*
	0.150	2.10	5.84	5.98	6.51	4.82	5.93	1.79	*4.71*
70.0	0.113	3.60	6.00	6.11	6.69	5.51	6.09	4.03	*5.43*
	0.130	4.85	6.20	6.40	6.95	5.74	6.25	4.43	*5.83*
	0.150	2.12	6.20	6.47	6.98	5.92	6.27	1.98	*5.13*
				With PV Systems and Battery ($\mathscr{D}$)					
40.0	0.113	0.81	9.93	10.03	10.42	9.47	9.33	0.82	*7.26*
	0.130	3.71	10.67	11.94	11.74	9.24	10.32	4.03	*8.81*
	0.150	1.79	11.48	13.28	12.78	8.57	11.20	1.76	*8.70*
50.0	0.113	0.88	8.51	8.90	9.09	7.38	7.69	0.94	*6.20*
	0.130	2.95	9.42	8.94	10.23	7.09	8.76	2.90	*7.18*
	0.150	1.33	9.61	8.52	10.65	6.63	9.03	1.23	*6.71*
60.0	0.113	0.75	6.77	7.10	8.04	6.39	6.45	0.82	*5.19*
	0.130	2.30	6.37	6.71	8.10	6.13	6.19	2.05	*5.41*
	0.150	0.92	5.79	6.17	7.71	5.79	5.61	1.11	*4.73*
70.0	0.113	0.71	5.00	5.15	6.09	5.30	4.74	0.77	*3.97*
	0.130	1.65	4.66	4.94	5.76	5.18	4.52	1.47	*4.03*
	0.150	0.73	4.27	4.54	5.32	4.84	4.10	0.64	*3.49*

supply. If this is scaled up to be the total grid load, then no baseload generation is required for those 4 h of the day. Practicalities of operating baseload plant mean these plants cannot just be switched off for 4 h. In actuality, the plant would be shut down permanently, and a more expensive gas-fired source would likely "fill the gap" left from 0:00 to 10:00 and again from 14:00 to 23:59. Consequently, the grid retail price is likely to increase substantially with losing the cost-benefit of access to low-cost baseload generation. Such an extreme undesirable scenario can be avoided by imposing regulatory policies, for example, a negative feed-in tariff and P2P clearing price, to discourage solar energy trading between 10:00 and 14:00. Therefore, a certain level of baseload generation is maintained with the household rooftop solar energy saved to batteries for night-time usages.

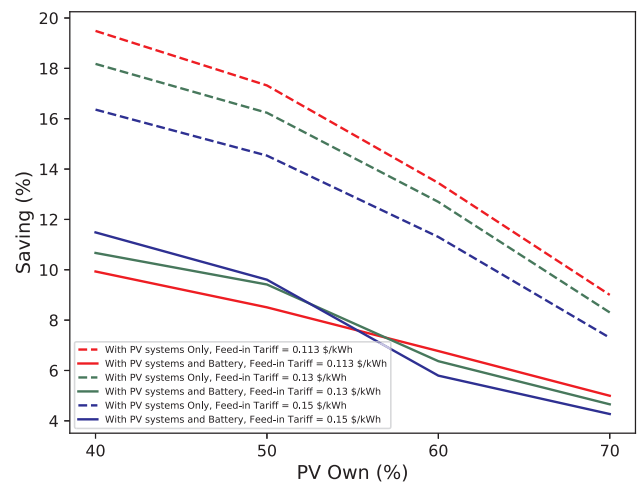

Fig. 10. Saving (%) of groups $\mathscr{B}$ and $\mathscr{D}$ – Feed-in tariff vs PV penetration.

5. Conclusion

This paper presents an optimization model for rooftop PV distributed generation with battery storage in P2P energy trading environment. The model is illustrated in a simulation framework for a local community with 500 households under real-world constraints which encompass PV systems, battery storages, customer demand profiles and market signals including retail prices, feed-in tariff, and P2P trading mechanism. The generated insights enable us to investigate complex interactions among P2P trading stakeholders under various scenarios, therefore shedding light on renewable energy policy-making and consumers/prosumers decision-making. These insights quantify the household savings under different scenarios of distributed renewable energy generation. Household energy savings are identified to be sensitive to many factors including the scale of PV systems, PV penetration, the peer-to-peer trading margins, the presence of battery storage and energy trading time. The model shows maximal savings up to 28% can be achieved by prosumers equipped with larger PV systems and battery storages during weekdays from an exemplified case. The sensitivity analysis demonstrates that households with PV systems have lower saving when PV penetration is high due to excessive energy traded on P2P market pushing down the clearing price and the saving gains. It is also shown that households with battery-only configuration achieve fewer benefits compared to normal households without any renewable resources in a P2P trading community. The insights mentioned above represent a starting point of the understanding of the impacts of policy, marketing and technical signals on economic benefits of household distributed renewable generation in a P2P energy trading market.

The following research works remain for us to extend. Developing an improved demand profile to represent real-world household energy consumption patterns better is a key step to approach economic feasibility of residential PV-battery DG. Advanced data analytics can be applied to model dynamic behaviors like human activities and weather events. This study is based on a basic battery model. In our opinion, higher fidelity models considering battery dynamics can enhance the technical resolution of the results. However, it may not be able to alter the final saving results as we found that batteries do not play a critical role in P2P energy trading due to their high operational costs. It is recommended to examine the techno-economic effects of a higher fidelity battery model on the optimized P2P energy trading of residential PV-battery DG. Owing to the complexity of practical constraints, a more

sophisticated method such as ADMM will need to be developed to ensure that optimal or near-optimal solutions can be obtained for large-scale instances within a reasonable runtime. Distributed optimization is also a challenging and interesting issue to be investigated to improve the scalability and the privacy of the optimization tool.

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